

Search Table and Annotated Bibliography

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Research Question: How will the early expiration of Novo Nordisk's Canadian patent for semaglutide-based drugs (Ozempic and Wegovy) affect the price and quantity of diabetes medication within Canadian brand and generic pharmaceutical markets?

Databases Search Tables

Table 1: EconLit Search Table

Search Terms			Date of Search	Number of Results	Results After Filters, Title Screening, Duplicate Removal*	Final Results After Abstract Screening**
Pharmaceutical			24/09/2025	1,658	9	4
► AND	Market Structure		24/09/2025	760		
► AND		Canada	24/09/2025	19		
Competitive Fringe Model			24/09/2025	29	7	5
► AND	Pharmaceuticals		24/09/2025	0		
Canadian Pharmaceutical Industry			24/09/2025	9	7	4
Drug price control OR Drug price ceiling			24/09/2025	32		
► AND	Canada		24/09/2025	2	13	9
Patented Medicine Prices Review Board			24/09/2025	4		
Pharmaceutical monopoly			24/09/2025	23	9	5
► AND	Property rights		24/09/2025	3		
► AND	Government-created		24/09/2025	0		
► AND	Canada		24/09/2025	0		
Generic Drug Market			26/09/2025	61	30	25
► AND	Canada		26/09/2025	0		
Generic Drug Entry			26/09/2025	54	15	12
► AND	Canada		26/09/2025	0		
Generic Drug Price			26/09/2025	89	14	7
► AND	Canada		26/09/2025	3		
Generic Drug Competition			26/09/2025	58	6	6
► AND	Canada		26/09/2025	1		
Brand drug			27/09/2025	132	4	3
► AND	Price		27/09/2025	34		
Drug price elasticity			27/09/2025	36	3	2
Diabetes drug			27/09/2025	24		
► AND	Demand		27/09/2025	3	2	1
► AND	Price		27/09/2025	3		
Drug patent			27/09/2025	232	6	4

► AND	Expiration		27/09/2025	23		
► AND		Canada	27/09/2025	2		
Ozempic OR Semaglutide			27/09/2025	4	0	0

* At this stage, filters, title screenings, and duplicate removals were applied to all search results. EconLit filters included: “English”, “Academic Journals”, “Dissertations”, and “Books”. Title screening criteria differed for each search term individually, but some overarching conditions included: no country-specific international case studies (with the exception of some in the U.S. and Europe), no mention of illicit drug markets, or no idiosyncratic historical case studies. Duplicate titles from all other searches in the table beforehand were removed.

** At this last stage, the abstracts of all remaining results from the previous column were screened. Again, the criteria was case dependent. Broadly, abstracts should have focused on the impacts of the search term on market structure, prices, supply, or demand. Alternatively, they could provide explanations and descriptions of the search term (for example, how the generic drug market operates). Abstracts were rejected due to focus on social consequences and disparities, U.S. insurance markets, U.S.-specific policy, health impacts of the drugs, outdated analysis, etcetera.

Table 2: Scopus Search Table

Search Terms			Date of Search	Number of Results	Results After Filters, Title Screening, Duplicate Removal ◇	Final Results After Abstract Screening ◇◇
Pharmaceutical			28/09/2025	561,320	3	0
► AND	Market Structure		28/09/2025	2,379		
► AND		Canada	28/09/2025	24		
Competitive Fringe Model			24/09/2025	230	8	2
► AND	Pharmaceuticals		28/09/2025	1		
Canadian Pharmaceutical Industry			28/09/2025	387	6	3
Drug price control OR Drug price ceiling			28/09/2025	5,130		
► AND	Canada		28/09/2025	197		
Patented Medicine Prices Review Board			28/09/2025	36	7	4
Pharmaceutical monopoly			28/09/2025	417		
► AND	Property rights		28/09/2025	77	5	2
► AND		Canada	28/09/2025	2		
► AND	Government-created		28/09/2025	0		
► AND	Canada		28/09/2025	10		

Generic Drug Market			28/09/2025	3,440	8	2
► AND	Canada		28/09/2025	128		
Generic Drug Entry			28/09/2025	753	4	0
► AND	Canada		28/09/2025	43		
Generic Drug Price			28/09/2025	2,888	2	0
► AND	Canada		28/09/2025	120		
Generic Drug Competition			28/09/2025	1,564	6	3
► AND	Canada		28/09/2025	54		
Brand drug			28/09/2025	11,521	12	12
► AND	Price		28/09/2025	1,125		
► AND	Canada		28/09/2025	46		
Drug price elasticity			28/09/2025	362	5	1
Diabetes drug			28/09/2025	383,190		
► AND	Demand		28/09/2025	2,335	0	0
► AND	Canada		28/09/2025	23		
► AND	Price		28/09/2025	869		
► AND	Canada		28/09/2025	20		
Drug patent			28/09/2025	36,864	7	6
► AND	Expiry		28/09/2025	366		
► AND	Canada		28/09/2025	18		
Ozempic OR Semaglutide			28/09/2025	7,193	0	0
► AND	Price		28/09/2025	78		

◇ At this stage, filters, title screenings, and duplicate removals were applied to all search results. Scopus filters included: “English”, “Academic Journals”, “Book Chapters”, and “Economics, Econometrics, and Finance”. Title screening criteria differed with each search term individually, but some overarching conditions included: no specific international case studies (with the exception of some in the U.S. and Europe), no mention of illicit drug markets, or no idiosyncratic historical case studies. A significant number of titles were also rejected for being cost benefit analyses of drug doses. Duplicate titles from all other searches in the table beforehand, in addition to all previously screened titles from EconLit, were removed. For a few topics, removal of screened EconLit studies eliminated almost all papers at this stage.

◇◇ At this last stage, the abstracts of all remaining results from the previous column were screened. Again, the criteria was case dependent. Broadly, abstracts should have focused on impacts of the search term on market structure, prices, supply, or demand. Alternatively, they could provide explanations and descriptions of the search term. Abstracts were rejected due to focus on social consequences and disparities, U.S. insurance markets, U.S.-specific policy, health impacts of the drugs, outdated analysis, etcetera. A few papers were rejected since they appeared merely as critiques of capitalism, with emphasis on corruption in pharmaceuticals.

Most Relevant Papers (ranked in order of relevance)

1. Duflos, Gautier, and Frank R. Lichtenberg (2011)

Duflos and Lichtenberg explore how changing market structure, primarily the loss of exclusivity through U.S. patent expiry, will affect market prices, the magnitude of marketing expenditures from pharmaceutical firms, and the quantity of drugs that will thereafter be consumed. They hypothesize that mean price and marketing expenditure are both inversely related to the generic firms' market share, while pharmaceutical utilization will transitively decrease with price and increase with the scope of marketing campaigns. Using monthly longitudinal, molecule-level data from 2000-2004, for over 19,000 prescription drugs sold in the U.S., Duflos and Lichtenberg conduct a series of log-linear regressions with fixed effects and fixed time effects. Results found that there is little generic competition in the first 12-years of a drug product's cycle, and then competition rises starkly from 10% to 54% on average within the next four years. This corresponds with an average 11.5-year period of patent protection in the U.S. During these same four years of generic entry, brand marketing expenditure also fell by over 50% on average. Relative price initially rose by 44% from years 0 to 12, and fell by around 61% from years 10 to 17. Hence, these two factors put opposing pressures on drug utilization, resulting in minimal change observed over the period after patent expiration – the increased demand due to reduced prices and the fall in demand due to less aggressive marketing offset each other's impact on total consumer surplus.

This study is highly relevant to my analysis, as it statistically models the evolution of pharmaceutical prices over their lifespan, both before and after patent expiration. There is clear evidence of increased generic market share and lowering prices, coinciding with what I have hypothesized will occur in the Canadian diabetes market once the semaglutide patent expires. It is also important to note that welfare may not change drastically, contrary to my initial assumption.

2. Nguyen, Nguyen Xuan , Steven H. Sheingold, Wafa Tarazi, and Arielle Bosworth (2022)

This journal article highlights the impact that loss of exclusivity and resulting generic competition in the pharmaceutical industry has on the average price of drugs. Data was employed from Medicare Part D claims, for years 2007-2018 and for 299 drugs each with at least two years of generic competition. Baseline prices are those of the brand name 6-months before entry of the first generic. A series of regression models with fixed and random effects were conducted, foremostly measuring the relationship between relative prices (standardized price of generics divided by price of brand before loss of exclusivity) and the number of generic producers in the market. Results elucidate that relative prices continue to fall monotonically, with each new entrant, over the second and third years following the first generic entry. In a market with around three competitors, relative prices were estimated to have fallen by 20% by the end of the second year, while a market with over ten competitors had estimated relative price decreases of up to 70%. Overall, more generic competition is associated with lower drug prices; the model suggests each entry could potentially reduce relative market price by 6%-8%.

The study closely mirrors my research question, but from an American perspective, and covering a stream of different drugs. It explicitly quantifies the potential decreases in pharmaceutical prices following patent expiry, which could in turn be useful as a rough numerical point of comparison for my model.

3. Frank, Richard G., and David S. Salkever (1992)

Frank and Salkever discuss the mechanisms whereby entrance of generic drug competition will consequently allow existing brand name drugs to increase their prices. They separate demand into two categories: hospitals and Medicare program holders, who are paid under reimbursement and hence are

highly elastic in demand for generics, and individuals purchasing through physician prescriptions, who are less elastic. The study models the industry using a Stackelberg game: the brand drug producer is the dominant firm who incorporates the price responses of the fringe generic producers into its profit maximization decisions. These “n” generic drug firms are modelled under a Nash equilibrium and must take the brand price as a given. All firms are faced with the two forms of demand. This model is delineated mathematically, with a few key observations drawn: as “n” increases, the price of generics will necessarily fall, and for there to be a rise in the brand drug price, the demand curve would need to steepen (signaling inelasticity based on brand loyalty or lack of information for example). The model is extended to include strategic marketing practices from the brand producer. Finally, the report undergoes a brief literature review, highlighting that there is not sufficient evidence to support that brand drug prices must fall during generic entry. The consequences for brand prices are not categorical; however, rising prices are feasible under this model and coincide with real-world observations in the U.S.

Despite being older, this is one of the few reports that directly connects the pharmaceutical industry and patent expiry to a specific Industrial Organization model. The Stackelberg with fringe appears to be a solid starting point for developing a more detailed model of diabetes drugs in Canada. Additionally, it is worth considering whether there is a possibility that brand Ozempic will raise its prices in the short run.

4. Vondoros, Sotiris, and Panos Kanavos (2013)

Vondoros and Kanavos explore the notion of the “generics paradox” – a phenomenon where the initial monopolistic producers of a drug increase their price after generics begin increasing competition – in the context of the European market. This differs from studies in the U.S. due to stricter regulation in Europe, including price ceilings and reference prices for generics, both of which widen the price gap between brand-name and generic drugs. The drug originators, who lose monopoly power with their patent expiry, are still able to maintain a sizeable market share even after competitive entry. This is primarily due to customer loyalty and price inelasticity for pharmaceuticals, which results in brands increasing their prices, thereby growing profits. This study uses pharmaceutical sales data from 1997-2002 for 6 European countries, and panel regression analysis, to estimate the relationship between the price of the originator and dummy variables for both the level of competition (generic price and generic market share) and the regulatory policy. Results suggest that prices of originators are around 3%-5% higher in markets with generic competitors – this evidence supports the existence of the generic paradox. Further, originator prices in markets with generic price ceilings experienced a further 2.2% higher price than those without such policies. It is proposed that price ceilings increase the demand for generics, but leave behind the most loyal brand customers with the highest inelasticity of demand, allowing the brand price to rise.

The price ceiling policies in Canada more closely align with those in Europe than in the U.S.; hence, in considering Novo Nordisk’s price setting after generic entry, this study will be highly relevant. It is possible that, in conjunction with Frank and Salkever’s analysis above, this same generic paradox may be observed with respect to brand Ozempic prices in Canada, particularly due to strong existing loyalty to it.

5. Granier, Laurent, and Sébastien Trinquard (2012)

Granier and Trinquard extend existing patent expiry models to include pseudo-generics, a drug launched by the brand company that is identical in composition to the original drug, but marketed as a generic. This can be a strategically beneficial tactic for the brand firm, and is often used as a method to acquire other generics, thereby reducing competition. The study represents this market structure as a Cournot model between the brand and generic firm, with imperfect substitutes and higher elasticity of demand for generics. During each period, the brand firm can acquire the generic, and within the first period, the brand also has the opportunity to launch a pseudo-generic. If it chooses to do so, then it becomes the follower in

a Stackelberg model for the generic drug market. Since competition lowers prices in generic firms, this makes it easier for the brand firm to acquire its generic competitors. While the launch of a pseudo-generic may reduce the brand firm's profit in the short run, the present value of its gain after acquisition may be significantly higher. Within the model, acquisition of the generic yields monopolistic benefits for the brand firm thereafter. The study concludes that it is not generally optimal for a brand firm to engage in the merger within the first period, and that pseudo-generics may successfully be launched with the sole intent of acquiring competitors.

Pseudo-generics compose a notable portion of the Canadian generic drug market, as stated in the study, and hence should be considered within analysis of the Ozempic patent expiration. It is possible that Novo Nordisk could launch a pseudo-generic, which can perhaps be included as an extension to my analysis.

6. **Bhattacharya, Jayanta, and William B. Vogt (2003)**
7. **Grabowski, Henry G., and Margaret Kyle (2007)**
8. **Kong, Ying (2004)**
9. **Ito, Yuki, Konan Hara, and Yasuki Kobayashi (2020)**
10. **Starc, Amanda, and Thomas G. Wollmann (2025)**

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